

Stochastic Simulation of Daily Precipitation and Daily Mean Temperatures

L. L. Golubyatnikov

*Oboukhov Institute of Atmospheric Physics, Russian Academy of Sciences,
Pyzhevskii per. 3, Moscow, 119017 Russia*

e-mail: golub@ifaran.ru

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Abstract—A method of simulating daily precipitation and daily mean surface air temperature is considered. The proposed algorithm is based on the simulation of dry and wet periods and on the method of inverse functions. The method is used to generate model time series of daily values of the precipitation amount and mean air temperature on the basis of data from weather stations located in different natural climatic zones. The time series of these climatic parameters simulated by the proposed stochastic generator are statistically adequate to the observed data series.

INTRODUCTION

Some processes occurring in the Earth's climate system have characteristic times that are much larger than the lengths of the observed time series. In studying such processes, the problem arises concerning the simulation of long time series of the parameters of these processes from available observational data, because the real time series are not long enough for a reliable estimation of the probability of occurrence of a given process. Model time series are also required to simulate meteorological conditions when their statistical characteristics change. These problems are solved using stochastic weather models, also called weather generators.

Weather generators are most often used to forecast major agroclimatic consequences of global warming [1, 2], to assess the influence of possible climate changes on land surface hydrology and water resources [3–5], and to analyze the response of vegetation to probable changes in the climatic parameters [6–8]. The problem of supplementing experimental time series with model values arises in the development of empirical climate models [9], in analysis of climatic variability under conditions of a changing climate [10], in study of the impact of global climatic changes on regional climates [11, 12], and in a risk assessment of carbon sequestration projects [13].

At present, several weather generators have been constructed to simulate model time series of climatic parameters from available data of weather station observations. Short surveys of stochastic weather models are given in [2, 11, 14]. Among the conceptual bases of these generators, two major approaches are noteworthy. The idea of one of them is that the lengths of sequences of consecutive dry or wet days (periods) alternating within the year are modeled using long-term observational data [15]. Dry days are those with no or

little precipitation, and wet days are with measurable precipitation. Then, climatic parameters, such as the amount of precipitation, temperature, solar radiation, and vapor pressure, are determined for each day of the indicated period depending on the length of a period and a specified distribution of the climatic parameters. The LARS-WG weather generator is based on this method [14]. In the second simulation method, a weather process is regarded as a first-order Markov chain with two states: dry days and wet days. The conditional transition probabilities are derived from the local weather data. Climatic parameters are simulated depending on the state and a given type of distribution. This approach forms the basis of weather generators such as WGEN [16], Met&Roll [2], CLIMGEN [17], and GEM [18]. It should be noted that stochastic weather models are usually local models and require a calibration with respect to observed data for each geographic location.

For simulating precipitation amounts, many of the weather generators employ a gamma distribution. This is related to the use of a minimum number of easily interpreted parameters of this distribution providing a fairly good approximation of observed data [19, 20]. In several works, precipitation amounts in the region are modeled using other special distribution laws, such as Pearson, Weibull, log-normal, mixed geometric, exponential, and mixed exponential distributions [15, 19, 20]. It should be noted that often a reliable estimate of the parameters of these distributions is difficult to obtain from available data. The values of air temperature are usually modeled by a normal distribution [15, 21]. However, some publications present the temperature distributions obtained by processing long-term meteorological series that are non-Gaussian [21, 22]. Thus, the often-used hypothesis that distribution functions for both precipitation and temperature are inde-

pendent of the geographic location is quite controversial.

The goal of this work is to develop a simulation method for daily precipitation amounts and daily mean surface air temperatures that is a priori independent of either any certain preliminarily specified types of distributions or any particular geographic location.

SIMULATION METHOD

In [11, 14], results are given of a comparison of the climatic series produced by weather generators based on the simulation of dry and wet periods and on the Markov approach. These studies have shown that weather generators based on the simulation of the lengths of dry and wet periods generate model series of climatic parameters that are more adequate to observed data. The stochastic weather model discussed here is based on the algorithm of analysis of the lengths of dry and wet periods alternating during the year.

The first step in solving the formulated problem is to find the distribution of the lengths of dry and wet periods for each month at a given geographic location from empirical data. For this, statistical series of the lengths of periods of the two given types are constructed using time series of observational data.

In analyzing period lengths, it is assumed that the period belongs to the month in which it has started. The length of a period is defined as the number of days for which the period lasted. Let us denote the number of periods of a given type of length j for the i th month ($i = \overline{1, 12}$) by k_j^i , the number of periods of this type in the given month by D^i , and the largest length of the period in this month by L^i . From k_j^i , we obtain the frequencies of periods of different lengths in the given month

$$p_j^i = k_j^i / D^i.$$

The set of j and corresponding frequencies p_j^i define the statistical series of a given period type for the i th month.

From the statistical series of the lengths of dry and wet periods for the i th month, it is possible to determine the distribution functions of the lengths of these periods. The empirical distribution functions of these parameters at the points $j = \overline{0, L^i}$ are specified as

$$F(0) = 0,$$

$$F(j) = a_j^i, \quad j = \overline{1, L^i - 1},$$

$$F(L^i) = 1,$$

where $a_j^i = \sum_{k=1}^j p_k^i$. Connecting the adjacent points (j, a_j^i) and $(j+1, a_{j+1}^i)$ by a broken line through the

point $(j+1, a_j^i)$, we obtain an approximate empirical stepwise distribution function for period lengths. Thus, the distribution functions used here for the lengths of dry and wet periods are written as

$$F(x) = \begin{cases} 0 & x < 1 \\ a_j & j \leq x < j+1, \quad j = \overline{1, L-1} \\ 1 & x \geq L. \end{cases} \quad (1)$$

The next step in solving the problem is to find the distributions of daily precipitation amounts and daily mean temperatures for each month of the year. Let $x_{\min}^i$ and $x_{\max}^i$ denote the minimum and maximum values of a given climatic parameter for the i th month ($i = \overline{1, 12}$). Let N^i denote the number of measurements of this parameter in the given month. The range of variation of the given parameter is partitioned into n equal parts $[x_{\min}^i, x_{\max}^i] = \bigcup_{j=1}^n [x_j^i, x_{j+1}^i]$, and the number of measurements of this parameter (m_j^i) is counted for each j th interval of this partitioning. From m_j^i , we obtain the occurrence frequencies for the corresponding intervals

$$w_j^i = m_j^i / N^i.$$

The set of intervals $[x_j^i, x_{j+1}^i]$ and corresponding frequencies w_j^i describe the statistical series of the analyzed climatic parameter for the i th month.

Suppose that the values of the climatic parameter for the i th month are unlikely in the intervals $[x_1^i = x_{\min}^i, x_k^i] = \bigcup_{j=1}^{k-1} [x_j^i, x_{j+1}^i]$ and $[x_t^i, x_n^i = x_{\max}^i] = \bigcup_{j=t}^{n-1} [x_j^i, x_{j+1}^i]$ if the frequency w_j^i in any subinterval $[x_j^i, x_{j+1}^i]$ of these intervals does not exceed $1/N^i$. The values of the climatic parameter for the i th month will be called regular if they lie in the interval $[x_k^i, x_t^i] = \bigcup_{j=k}^{t-1} [x_j^i, x_{j+1}^i]$.

There are not enough data to find an appropriate distribution of rare values of the given climatic parameters. In this study, a uniform distribution of these temperature and precipitation values is assumed as the simplest mathematical hypothesis.

From the statistical series of the climatic characteristics for the i th month, one can obtain distribution functions for their regular values. Without loss of generality, we assume that regular values, both precipitation and temperature, satisfy the equality $\sum_{j=1}^n w_j^i = 1$. The empirical distribution functions for the analyzed climatic parameters at the boundary points of the inter-

Table 1. Natural and climatic characteristic of geographic locations

Weather station	Köppen–Trewartha climate type	Natural climatic zone	Period
St. Petersburg (60° N, 30°21' E)	Temperate continental	Middle taiga	1900–1990
Magadan (59°32' N, 150°49' E)	Subarctic continental	Forest-tundra	1936–1990
Bezenchuk (53°12' N, 50°24' E)	Temperate continental	Temperate arid steppe	1955–1983
Orenburg (51°47' N, 55°08' E)	Temperate continental	Arid steppe	1900–1990
Irkutsk (52°28' N, 104°20' E)	Temperate continental	Southern taiga	1910–1990
Kamennaya Step (51°07' N, 40°50' E)	Temperate continental	Temperate arid steppe	1955–1986
Kompolt (47°42' N, 20°17' E)	Temperate maritime	Meadow steppe	1956–1985

vals $[x_j^i, x_{j+1}^i]$, which are present in the statistical series for these parameters, are specified as

$$F(x_1^i) = 0,$$

$$F(x_j^i) = b_j^i, \quad j = \overline{2, n-1},$$

$$F(x_n^i) = 1,$$

where $b_j^i = \sum_{k=1}^j w_k^i$. Connecting the adjacent points by a straight line, we obtain an approximate empirical distribution function for a given climatic parameter. Thus, the distribution functions used here for daily precipitation and daily mean temperatures are written as

$$F(x) = \begin{cases} 0, & x < x_1 \\ w_j \frac{x - x_{j-1}}{x_j - x_{j-1}} + b_{j-1} & x_{j-1} \leq x < x_j, j = \overline{2, n} \\ 1, & x \geq x_n. \end{cases} \quad (2)$$

Hence, the problem is to simulate random variables that fit the distribution functions (1) and (2). For this purpose, the method of inverse functions will be used. The method transforms the sequence of random numbers uniformly distributed on the interval (0, 1) into a sequence of random numbers with a given distribution function $F(x)$. A substantiation of the method can be found in [22]. According to the method of inverse functions, for the indicated transformation, one must select a random number y out of the set of random numbers uniformly distributed on the interval (0, 1) and solve the equation

$$F(x) = y$$

for x . The solution of this equation

$$X = F^{-1}(Y) \quad (3)$$

(where F^{-1} is the inverse of the function F) is a random number from the set of random numbers with the distribution function $F(x)$.

Thus, according to (3), the sequence of random numbers with distribution function (2) can be deter-

mined from the sequence of random numbers uniformly distributed on the interval (0, 1):

$$x = F^{-1}(y) = (x_j - x_{j-1})(y - b_{j-1})/w_j + x_{j-1} \quad (4)$$

if $b_{j-1} \leq y < b_j$ and $b_j = \sum_{k=1}^j w_k$.

If the sequence of random numbers has step distribution function (1), formula (4) can be written as

$$x = \min(j : y \leq a_j)$$

if $a_{j-1} \leq y < a_j$ and $a_j = \sum_{k=1}^j p_k$.

RESULTS

The actual data used in this study were daily mean air temperatures and daily precipitation from the following weather stations: Bezenchuk, Irkutsk, Kamennaya Step, Magadan, Orenburg, St. Petersburg (Russia), and Kompolt (Hungary). The stations are located between 47°–60° N in different natural climatic zones (Table 1).

Analysis of the observed and model values of the length of dry and wet periods has shown that the proposed model satisfactorily simulates real weather series. Figure 1 shows the actual and model monthly mean values of the length of dry and wet periods. It follows from the graphs that the mean length of wet periods for the indicated geographic locations varies from 1 to 4 days; the mean length of dry periods lies in the range from 2 to 8 days. As a rule, the mean lengths of dry periods in each month are larger than the lengths of wet periods during the year. Data for the northern taiga are an exception. Analysis of St. Petersburg weather station data shows that the mean length of wet periods from September to March is larger than the length of dry periods. Thus, it follows that the proposed method adequately describes the change in the length of dry and wet periods during the year. The differences between the observed and model values average no more than 6% for dry series and up to 8% for wet series.

The lengths and probabilities of occurrence of dry periods during the time interval under study are important in problems connected with the assessment of consequences of agroclimatic changes and with the forecast of changes in hydrologic processes. Figure 2 depicts the probability of occurrence of dry periods

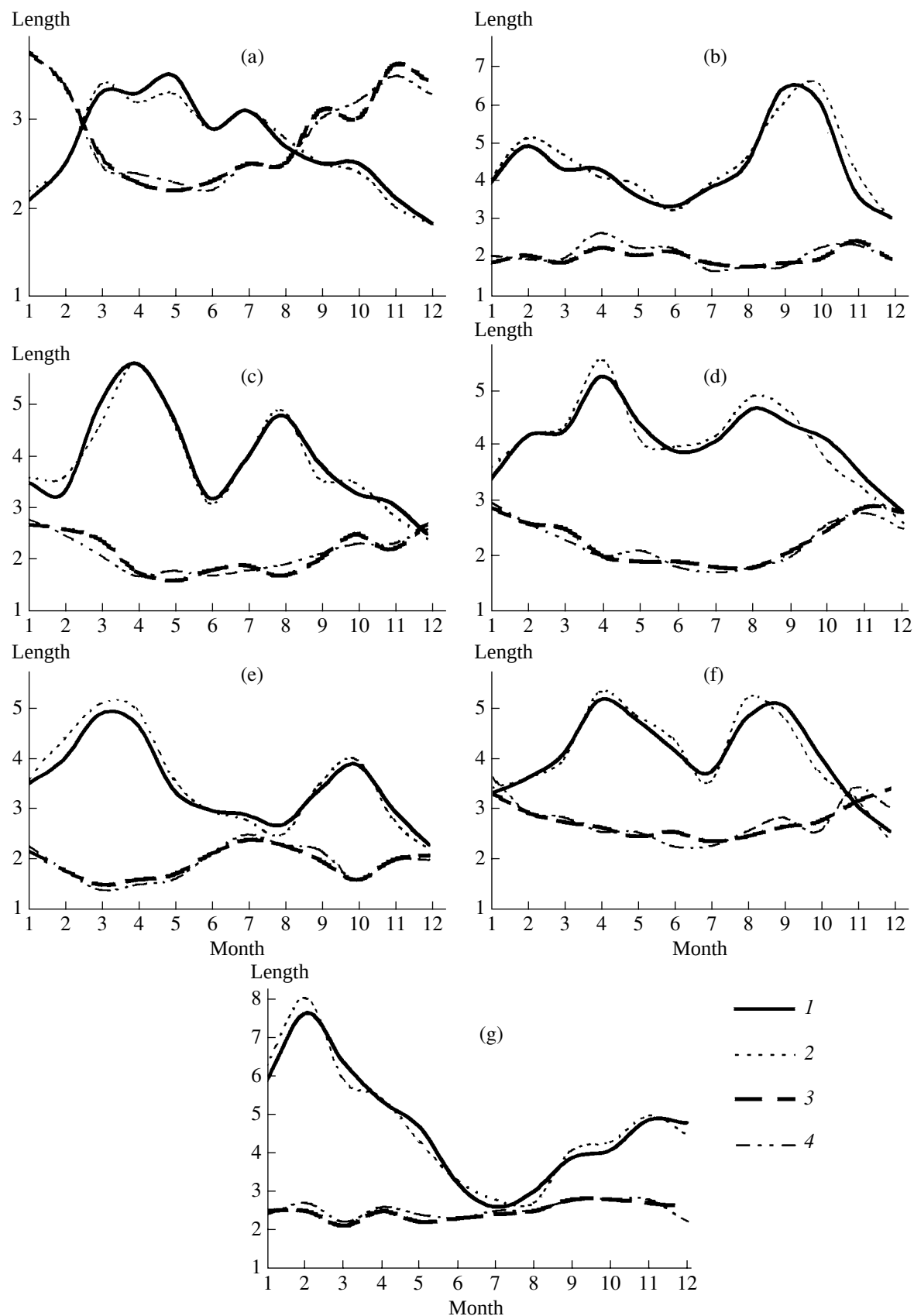


Fig. 1. (1, 3) Observed and (2, 4) model values of monthly mean lengths of (1, 2) dry and (3, 4) wet periods for (a) St. Petersburg, (b) Kompolt, (c) Bezenchuk, (d) Orenburg, (e) Irkutsk, (f) Kamennaya Step, and (g) Magadan.

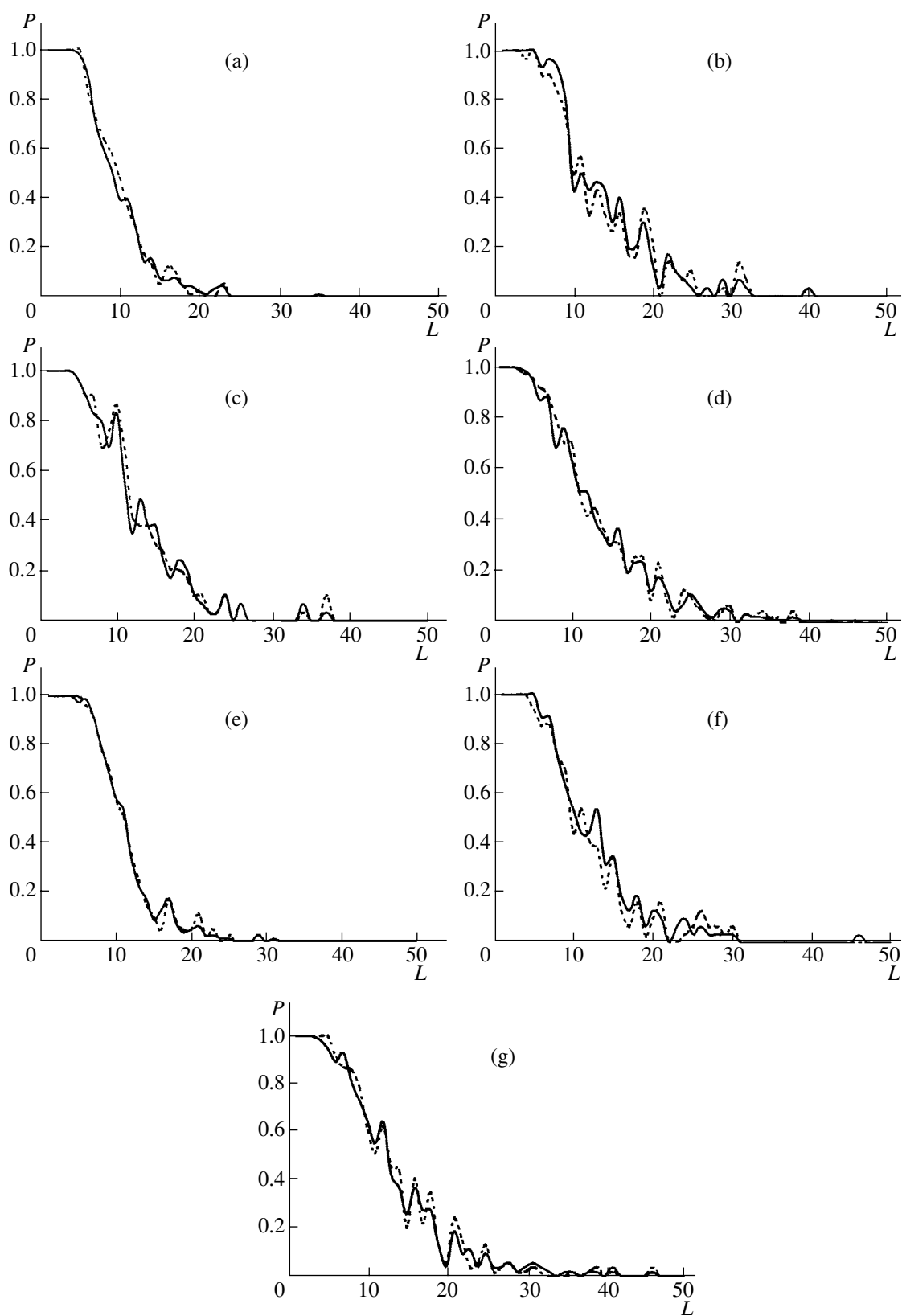


Fig. 2. Probability of occurrence (P) of a dry period of length L during the year from observed (solid line) and model (dashed line) data for (a) St. Petersburg, (b) Kompolt, (c) Bezenchuk, (d) Orenburg, (e) Irkutsk, (f) Kamennaya Step, and (g) Magadan.

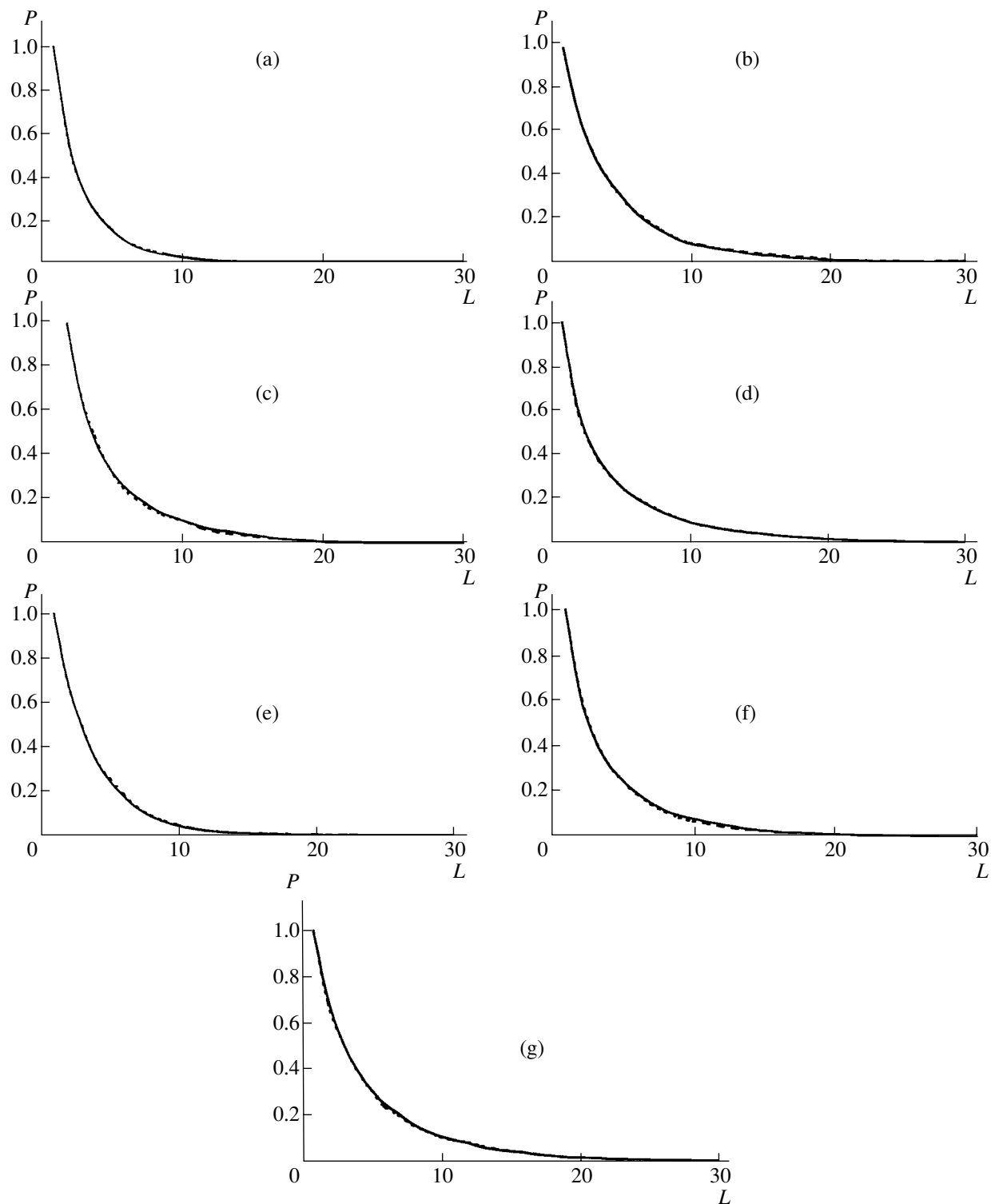


Fig. 3. Probability of occurrence (P) during the year of a dry period with a length greater than L from observed (solid line) and model (dashed line) data for (a) St. Petersburg, (b) Kompolt, (c) Bezenchuk, (d) Orenburg, (e) Irkutsk, (f) Kamennaya Step, and (g) Magadan.

with different lengths during a year from the observed and model data for the indicated geographic sites. It follows that the model adequately simulates the sharp drop in the probability of occurrence of long dry peri-

ods in the forest zone (Figs. 2a, 2e), a smoother decrease in the steppe zone (Figs. 2b–2d, 2f), and the probability of long dry periods in the Far East forest-tundra zone (Fig. 2g). The probabilities of occurrence

Table 2. Means ($\bar{x}$, mm) and standard deviations (σ) of actual (1) and model (2) daily precipitation in wet periods (estimate from n time series)

Weather station	Data		Month											
			Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
St. Petersburg $n = 91$	1	$\bar{x}$	1.7	1.8	2.1	2.7	3.6	4.4	4.9	5.2	3.9	3.4	2.5	1.9
		σ	2.4	2.2	2.6	3.4	4.8	5.3	6.5	7.7	4.8	4.0	3.1	2.4
	2	$\bar{x}$	1.7	1.8	2.1	2.6	3.6	4.3	5.0	4.9	4.0	3.4	2.6	1.9
		σ	2.3	2.2	2.8	3.1	4.9	5.3	6.4	6.5	5.1	4.1	3.2	2.4
Magadan $n = 55$	1	$\bar{x}$	1.6	1.6	1.4	3.0	3.5	4.2	4.3	5.0	6.3	5.4	3.6	2.3
		σ	2.2	2.3	3.0	6.4	5.6	6.6	7.0	9.0	9.1	8.2	5.7	3.4
	2	$\bar{x}$	1.7	1.7	1.4	3.4	3.4	4.4	4.1	5.1	5.7	5.2	3.7	2.3
		σ	2.4	2.4	3.1	6.9	5.7	6.9	7.0	9.6	8.3	8.1	6.5	3.3
Bezenchuk $n = 29$	1	$\bar{x}$	2.5	2.2	2.3	3.3	3.7	5.0	4.8	4.4	4.2	3.5	2.4	2.2
		σ	3.1	2.6	2.9	4.6	4.9	7.1	7.3	5.6	5.7	4.2	3.1	2.8
	2	$\bar{x}$	2.6	2.5	2.2	3.5	3.8	4.9	5.0	4.2	4.3	3.7	2.5	2.0
		σ	3.2	3.0	2.9	5.1	5.6	7.0	7.8	5.6	5.8	4.5	3.1	2.4
Orenburg $n = 91$	1	$\bar{x}$	1.5	1.6	1.7	2.8	3.7	4.0	4.1	3.7	3.4	3.1	2.4	1.7
		σ	2.3	2.2	2.6	4.1	5.2	5.7	5.9	5.7	4.7	4.0	3.2	2.3
	2	$\bar{x}$	1.6	1.6	1.8	2.7	3.6	3.8	4.2	3.7	3.4	3.1	2.4	1.7
		σ	2.3	2.2	2.8	3.9	5.1	5.8	6.1	6.2	4.6	3.8	3.2	2.4
Irkutsk $n = 81$	1	$\bar{x}$	0.9	1.0	1.4	2.2	3.1	5.8	7.3	6.1	4.0	2.5	1.4	1.1
		σ	0.9	2.7	1.9	3.0	4.0	8.5	10.1	9.0	5.8	4.1	1.7	1.4
	2	$\bar{x}$	0.9	0.9	1.4	2.2	3.1	5.8	6.7	6.0	3.8	2.8	1.4	1.1
		σ	1.0	2.1	2.0	3.2	4.1	8.5	9.6	9.3	5.3	5.3	1.8	1.5
Kamennaya Step $n = 32$	1	$\bar{x}$	2.0	1.8	1.8	3.3	4.3	4.8	4.3	4.5	4.0	3.2	2.6	2.1
		σ	2.8	2.5	2.4	4.3	5.7	6.8	6.4	6.4	6.1	4.3	3.4	3.0
	2	$\bar{x}$	2.2	1.9	1.8	3.2	4.4	4.7	4.4	4.8	3.9	2.9	2.8	2.1
		σ	3.0	2.8	2.3	4.1	5.8	6.8	7.0	6.8	6.3	3.9	3.6	3.1
Kompolt $n = 30$	1	$\bar{x}$	3.2	3.6	3.4	4.1	4.9	6.8	5.8	7.0	5.1	5.5	4.6	3.6
		σ	4.3	4.5	4.2	5.3	5.9	9.0	8.1	9.6	7.4	7.5	6.1	5.0
	2	$\bar{x}$	3.4	4.0	3.6	3.9	4.8	7.0	5.7	7.2	5.5	5.5	4.7	3.3
		σ	4.4	4.5	4.6	4.9	6.1	9.4	8.5	10.4	7.5	7.7	6.0	4.4

during the year of a dry period whose length is greater than the given value are shown for the stations of interest in Fig. 3. It is seen that deviations of the model probabilities from the observed values are not large. Thus, the proposed weather generator simulates well the indicated extreme weather characteristics.

The values of daily precipitation and daily mean air temperatures generated by the proposed simulation

method agree well with the observed data. The means and standard deviations of the observed and model daily precipitation in wet periods and daily mean air temperatures, calculated from the corresponding time series of daily data for each month, are given in Tables 2 and 3. It can be seen that differences between the model and observed values of these climatic characteristics lie in the range of measurement accuracy.

Table 3. Means ($\bar{x}$, °C) and standard deviations (σ) of actual (1) and model (2) daily mean air temperatures (estimate from n time series)

Weather station	Data		Month											
			Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
St. Petersburg $n = 91$	1	$\bar{x}$	-7.6	-7.5	-3.4	3.5	10.2	15.3	18.0	16.3	11.0	5.1	-0.2	-4.8
		σ	2.0	1.9	2.0	2.3	3.5	4.1	4.9	6.0	4.0	3.5	2.7	2.1
	2	$\bar{x}$	-7.6	-7.5	-3.4	3.6	10.1	15.2	17.9	16.3	11.0	5.1	-0.2	-4.6
		σ	2.0	1.9	2.1	2.4	3.6	4.0	5.0	5.0	4.2	3.5	2.8	2.1
Magadan $n = 55$	1	$\bar{x}$	-17.1	-15.9	-12.5	-5.5	1.3	6.9	11.2	11.7	7.1	-2.1	-11.0	-15.2
		σ	6.2	6.0	5.5	3.8	2.9	2.8	2.1	1.9	2.7	4.9	5.2	5.9
	2	$\bar{x}$	-17.0	-15.9	-12.5	-5.5	1.3	6.9	11.2	11.6	7.1	-2.2	-11.2	-15.4
		σ	6.3	6.1	5.4	3.8	2.8	2.8	2.2	1.9	2.7	4.8	5.3	5.8
Bezenchuk $n = 29$	1	$\bar{x}$	-12.2	-12.4	-5.9	6.7	15.3	19.0	20.7	19.1	12.8	4.7	-2.6	-8.7
		σ	7.8	7.8	6.6	5.5	4.7	4.4	3.5	3.9	4.7	4.8	5.1	7.0
	2	$\bar{x}$	-12.1	-12.4	-6.0	6.8	15.3	18.9	20.7	19.1	12.7	5.1	-2.7	-8.6
		σ	8.1	7.5	6.8	5.6	4.7	4.2	3.5	4.0	4.7	4.3	5.0	7.0
Orenburg $n = 91$	1	$\bar{x}$	-14.1	-13.9	-7.0	5.7	15.3	20.2	22.3	20.1	13.3	4.4	-3.8	-10.7
		σ	7.5	7.4	6.6	6.3	4.9	4.5	3.8	4.1	4.9	4.8	5.9	7.1
	2	$\bar{x}$	-14.0	-13.9	-7.1	5.9	15.2	20.2	22.1	20.2	13.4	4.4	-3.8	-10.6
		σ	7.6	7.3	6.6	6.3	4.9	4.5	3.9	4.2	4.9	4.8	6.0	7.1
Irkutsk $n = 81$	1	$\bar{x}$	-20.0	-17.5	-8.9	1.4	9.0	15.2	17.9	15.1	8.3	0.9	-9.9	-17.9
		σ	6.7	6.0	6.1	4.6	4.4	3.6	2.8	3.4	3.5	4.7	6.9	7.5
	2	$\bar{x}$	-20.0	-17.4	-9.0	1.5	9.0	15.1	17.9	15.0	8.4	0.9	-10.1	-17.9
		σ	6.9	5.9	6.1	4.4	4.4	3.6	2.7	3.4	3.6	4.8	6.9	7.5
Kamennaya Step $n = 32$	1	$\bar{x}$	-8.9	-8.8	-3.4	7.5	15.0	18.4	19.9	19.0	13.2	5.9	-0.9	-5.6
		σ	7.0	6.9	5.6	5.1	4.7	4.0	3.4	4.0	4.7	4.6	5.0	6.3
	2	$\bar{x}$	-9.0	-8.7	-3.4	7.3	15.2	18.2	19.9	18.9	13.6	5.8	-0.9	-5.3
		σ	7.3	6.7	5.7	5.1	4.6	3.9	3.5	4.0	4.5	4.5	5.0	6.2
Kompolt $n = 30$	1	$\bar{x}$	-2.2	0.3	4.7	10.7	15.6	19.3	20.9	20.3	16.2	10.5	4.9	0.2
		σ	4.5	4.3	4.3	3.8	3.7	3.4	3.2	3.2	3.6	3.8	4.1	3.9
	2	$\bar{x}$	-2.3	0.3	4.6	10.8	15.7	19.2	20.9	20.3	16.2	10.8	4.7	0.4
		σ	4.7	4.2	4.4	3.9	3.6	3.2	3.2	3.2	3.5	3.6	4.1	3.9

Table 4. Statistics (lower quartile $x_{0.25}$, median $x_{0.5}$, upper quartile $x_{0.75}$) of actual (1) and model (2) distributions of daily precipitation (mm) in wet periods

Weather station	Statistics	Month																							
		Jan.		Feb.		Mar.		Apr.		May		June		July		Aug.		Sept.		Oct.		Nov.		Dec.	
		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
St. Petersburg	$x_{0.25}$	0.3	0.3	0.3	0.3	0.3	0.4	0.5	0.4	0.6	0.5	0.8	0.8	0.7	0.9	0.9	0.1	0.7	0.7	0.6	0.5	0.4	0.4	0.3	0.4
	$x_{0.5}$	0.8	0.8	1.0	1.0	1.1	1.1	1.5	1.4	1.9	1.8	2.6	2.6	2.4	2.8	2.7	2.6	2.1	2.1	1.8	1.8	1.3	1.3	1.0	0.9
	$x_{0.75}$	2.1	2.2	2.5	2.5	2.7	2.8	3.6	3.6	4.8	4.9	5.9	5.6	6.4	6.9	6.7	6.6	5.3	5.5	4.8	4.7	3.6	3.6	2.6	2.3
Magadan	$x_{0.25}$	0.3	0.2	0.3	0.3	0.2	0.2	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.3	0.5	0.7	0.1	0.6	0.7	0.5	0.4	0.4	0.4
	$x_{0.5}$	0.7	0.8	0.7	0.6	0.6	0.6	1.0	1.1	1.5	1.5	1.5	1.5	1.4	1.3	1.4	1.4	2.7	2.4	2.3	2.2	1.7	1.6	1.0	1.0
	$x_{0.75}$	1.8	2.2	1.9	2.2	1.5	1.5	3.1	3.0	4.3	4.1	5.0	5.0	5.1	4.8	5.5	5.9	8.2	7.9	6.1	5.8	4.6	4.4	3.2	2.9
Bezenchuk	$x_{0.25}$	0.5	0.4	0.4	0.4	0.4	0.4	0.7	0.9	0.6	0.6	0.8	0.8	0.9	0.8	0.7	0.6	0.8	0.7	0.7	0.8	0.4	0.4	0.4	0.4
	$x_{0.5}$	1.3	1.5	1.1	1.2	1.3	1.2	1.9	1.9	1.7	1.6	2.5	2.5	2.3	2.4	2.1	2.1	2.2	2.2	2.2	2.3	1.3	1.4	1.2	1.1
	$x_{0.75}$	3.1	3.4	3.0	3.4	3.0	3.0	4.1	4.3	4.8	4.6	6.5	6.2	5.2	6.0	5.9	5.2	4.7	4.8	4.4	4.4	3.2	3.2	2.9	2.5
Orenburg	$x_{0.25}$	0.3	0.2	0.3	0.3	0.2	0.2	0.4	0.4	0.5	0.5	0.6	0.6	0.6	0.6	0.5	0.5	0.6	0.6	0.5	0.6	0.4	0.4	0.3	0.3
	$x_{0.5}$	0.7	0.8	0.8	0.8	0.8	0.8	1.3	1.3	1.7	1.6	1.7	1.6	1.8	1.8	1.7	1.7	1.6	1.6	1.6	1.7	1.1	1.1	0.8	0.8
	$x_{0.75}$	1.8	2.0	2.1	2.0	2.0	2.1	3.4	3.3	4.5	4.6	5.0	4.8	5.1	5.2	4.5	4.2	4.1	4.2	4.2	4.3	3.0	3.3	2.2	2.2
Irkutsk	$x_{0.25}$	0.3	0.1	0.2	0.2	0.3	0.3	0.5	0.4	0.7	0.6	0.7	0.8	1.0	0.5	0.7	0.7	0.6	0.6	0.4	0.6	0.3	0.3	0.3	0.3
	$x_{0.5}$	0.6	0.5	0.6	0.6	0.7	0.7	1.1	1.1	1.7	1.7	2.6	2.6	3.2	3.1	2.5	2.5	2.0	2.0	1.3	1.4	0.8	0.8	0.6	0.6
	$x_{0.75}$	1.2	1.3	1.2	1.2	1.6	1.7	2.7	2.7	4.2	4.0	7.3	7.4	9.7	9.3	8.0	7.9	5.0	4.7	3.2	3.4	1.9	1.8	1.3	1.4
Kamennaya Step	$x_{0.25}$	0.3	0.2	0.2	0.2	0.3	0.4	0.5	0.6	0.6	0.6	0.5	0.4	0.6	0.5	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.5	0.3	0.4
	$x_{0.5}$	0.9	1.1	0.8	0.8	0.9	0.9	1.6	1.8	2.1	2.2	2.0	2.0	2.0	1.8	1.9	2.0	1.8	1.8	1.6	1.6	1.2	1.3	1.0	1.0
	$x_{0.75}$	2.6	3.0	2.2	2.6	2.2	2.2	4.3	4.3	5.5	5.5	5.9	5.8	5.0	4.5	5.9	6.8	4.8	4.8	4.2	3.9	3.6	4.0	2.5	2.5
Kompolt	$x_{0.25}$	0.4	0.3	0.5	0.5	0.5	0.5	0.7	0.7	0.8	0.9	1.2	1.0	0.9	0.7	1.0	1.1	0.7	0.7	0.8	0.7	0.6	0.5	0.4	0.1
	$x_{0.5}$	1.6	2.0	1.8	2.2	1.8	1.8	2.1	2.1	2.5	2.6	3.6	3.7	3.0	2.8	3.5	3.4	2.3	2.5	2.4	2.7	2.3	2.2	1.7	1.7
	$x_{0.75}$	4.4	4.5	5.5	6.2	4.5	4.9	5.5	5.8	6.7	6.1	9.5	9.8	7.4	7.1	9.1	9.2	6.2	6.9	6.4	6.6	5.9	6.0	5.1	4.9

The degree of coincidence of the experimental and model distributions of daily precipitation and daily mean air temperatures can be estimated from statistical characteristics such as medians and lower and upper quartiles. These numerical characteristics of distributions give insight into the position and scatter of the values being studied. For the stations of interest, Tables 4 and 5 show the medians and lower and upper quartiles for each month obtained from the corresponding time series of daily data. From these tables, it follows that the distributions of the model and observed values of the climatic parameters agree well with each other. From Tables 3 and 5, it may be concluded that the distribution of daily mean air temperatures can in some cases be described quite satisfactorily by a normal law,

e.g., daily mean air temperatures in August for the Orenburg station, in May for Kamennaya Step, and in September for Kompolt. At the same time, it is easy to see that some of the distributions of this parameter are asymmetric. For example, in the distributions of daily mean air temperatures in November for the Orenburg station, in January for Kompolt, and in March for Bezenchuk, the means and the medians do not coincide, which contradicts the properties of a symmetric distribution.

Thus, the testing of the proposed method of simulation of daily mean surface air temperatures and daily precipitation, performed using the data of meteorological observations of these climatic parameters, has dem-

Table 5. Statistics (lower quartile $x_{0.25}$, median $x_{0.50}$, upper quartile $x_{0.75}$) of actual (1) and model (2) distributions of daily mean air temperatures ($^{\circ}\text{C}$)

Weather station	Statistics	Month																							
		Jan.		Feb.		Mar.		Apr.		May		June		July		Aug.		Sept.		Oct.		Nov.		Dec.	
		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
St. Petersburg	$x_{0.25}$	-12.2	-12.5	-11.8	-11.7	-6.5	-6.5	0.9	1.0	6.9	6.7	12.5	12.5	15.8	15.8	14.1	14.2	8.8	8.7	2.4	2.4	-3.0	-2.9	-8.2	-8.2
	$x_{0.5}$	-6.5	-6.3	-6.7	-6.7	-2.4	-2.5	3.1	3.2	10.0	9.9	15.1	15.1	17.8	17.7	16.1	16.1	11.1	11.1	5.3	5.3	0.3	0.4	-3.8	-3.6
	$x_{0.75}$	-2.0	-1.8	-2.6	-2.8	0.6	0.6	6.0	6.2	13.3	13.1	18.0	18.0	20.1	19.9	18.4	18.4	13.2	13.3	8.0	8.0	2.8	2.7	0.0	0.1
Magadan	$x_{0.25}$	-21.6	-21.7	-20.2	-20.5	-16.5	-16.2	-7.7	-7.7	-0.7	-0.7	5.0	5.1	9.8	9.9	10.4	10.3	5.5	5.4	-5.2	-5.3	-14.7	-14.7	-19.7	-19.9
	$x_{0.5}$	-17.8	-17.6	-16.0	-16.0	-13.0	-12.9	-5.1	-5.3	1.2	1.1	6.8	6.9	11.1	11.2	11.5	11.4	7.4	7.5	-1.6	-1.8	-11.1	-11.3	-15.6	-15.7
	$x_{0.75}$	-13.0	-12.8	-11.8	-11.7	-8.5	-8.5	-2.8	-2.9	3.2	3.2	8.7	8.7	12.4	12.5	12.7	12.7	9.0	9.1	1.6	1.4	-7.3	-7.5	-11.2	-11.5
Bezenchuk	$x_{0.25}$	-17.7	-18.2	-17.7	-17.6	-10.7	-10.9	2.7	2.6	11.9	12.0	16.0	16.0	18.3	18.3	16.3	16.3	9.4	9.3	1.8	2.5	-5.2	-5.4	-13.4	-12.8
	$x_{0.5}$	-11.2	-10.8	-12.3	-12.3	-4.7	-4.5	6.5	6.8	15.6	15.5	19.1	18.9	20.6	20.4	19.0	19.1	12.8	12.6	4.8	5.3	-1.5	-1.6	-7.7	-7.6
	$x_{0.75}$	-6.2	-5.6	-6.0	-6.2	-0.4	-0.3	10.7	10.8	18.9	18.8	22.0	21.8	23.2	23.2	21.7	21.8	16.0	15.5	7.8	8.1	0.7	0.7	-2.9	-2.9
Orenburg	$x_{0.25}$	-19.4	-19.7	-19.3	-19.2	-11.8	-11.8	1.4	1.7	11.8	11.7	17.3	17.2	19.7	19.6	17.2	17.3	9.8	9.8	1.4	1.4	-7.0	-7.0	-15.6	-15.6
	$x_{0.5}$	-13.8	-13.6	-14.3	-14.5	-6.6	-6.7	5.4	5.7	15.6	15.6	20.5	20.3	22.2	22.1	20.1	20.2	13.4	13.4	4.4	4.3	-2.8	-2.7	-10.0	-9.9
	$x_{0.75}$	-8.4	-8.2	-8.1	-8.4	-1.9	-2.0	10.3	10.6	18.9	18.8	23.6	23.6	24.9	24.7	23.0	23.2	16.5	16.8	7.6	7.7	0.3	0.3	-5.2	-5.0
Irkutsk	$x_{0.25}$	-24.1	-24.4	-21.5	-21.3	-12.8	-12.8	-1.3	-1.1	5.8	5.9	12.7	12.6	16.2	16.2	12.8	12.8	5.9	5.9	-1.8	-1.8	-14.3	-14.4	-22.6	-22.6
	$x_{0.5}$	-19.4	-19.2	-17.3	-17.1	-8.3	-8.5	1.5	1.6	8.8	8.7	15.4	15.3	18.0	17.9	15.3	15.2	8.5	8.6	1.2	1.2	-9.1	-9.3	-17.1	-17.0
	$x_{0.75}$	-15.4	-15.3	-13.4	-13.3	-4.4	-4.5	4.5	4.4	12.1	12.2	17.8	17.8	19.9	19.7	17.5	17.4	10.7	10.9	4.0	4.1	-4.8	-4.8	-12.3	-12.3
Kamen-naya Step	$x_{0.25}$	-14.3	-15.0	-14.3	-14.0	-7.0	-7.1	3.3	3.2	11.4	11.9	15.6	15.4	17.4	17.4	16.2	16.2	9.7	10.5	3.0	2.9	-3.9	-3.7	-9.3	-8.6
	$x_{0.5}$	-7.9	-7.8	-8.4	-8.1	-2.4	-2.2	7.5	7.3	15.0	15.2	18.4	18.2	19.7	19.6	18.7	18.7	13.2	13.6	5.9	5.9	-0.2	-0.1	-4.3	-4.1
	$x_{0.75}$	-3.0	-2.5	-2.8	-2.8	0.4	-0.4	11.5	11.2	18.6	18.9	21.1	20.9	22.3	22.3	21.8	21.4	16.6	16.8	9.2	9.2	2.5	2.6	-0.9	-0.8
Kompolt	$x_{0.25}$	-4.8	-5.4	-2.0	-1.8	1.7	1.4	7.9	8.0	13.1	13.3	17.0	16.8	18.7	18.7	18.0	18.0	13.6	13.8	7.7	8.3	1.9	1.9	-2.1	-2.0
	$x_{0.5}$	-1.4	-1.2	0.7	0.8	4.6	4.7	10.5	10.6	15.7	15.8	19.4	19.2	21.0	20.8	20.1	20.1	16.2	16.2	10.3	10.8	5.0	4.7	0.3	0.4
	$x_{0.75}$	1.1	1.3	2.9	2.8	7.7	7.8	13.3	13.2	18.2	18.2	21.8	21.7	23.2	23.1	22.4	22.5	18.8	18.6	13.2	13.5	8.0	7.9	2.8	3.0

onstrated a good consistency between the empirical data and model calculations.

CONCLUSIONS

The proposed weather generator is based on analysis of the climatic parameters from the viewpoint of modeling a sequence of dry and wet periods. The stochastic generator of daily precipitation and daily mean surface air temperature is a priori not related to any certain preliminarily specified types of distributions or any particular geographic location. This allows us to suggest that the proposed method can be successfully used to generate time series of the indicated climatic parameters from observational data in any geographic region. The method has been tested on data from seven weather stations located in different natural climatic zones.

Comparison of the experimental and model estimates for the lengths of dry and wet periods and for the probability of occurrence of dry periods with different lengths has shown that the model adequately simulates the dynamics of the lengths of dry and wet periods.

Analysis of statistical characteristics, such as means, standard deviations, medians, and lower and upper quartiles, for distributions of daily precipitation amounts and daily mean surface air temperatures shows a good consistency of the model estimates and actual values of the climatic parameters being studied. The proposed method of simulating daily precipitation and daily mean temperatures generates the series of these climatic parameters that are statistically adequate to the original data.

Thus, the proposed method provides a fairly good approximation of the climatic parameters to empirical distributions. The results are indicative of the reliability of this method and give grounds to suggest that the proposed stochastic model can be successfully used in prognostic ecoclimatic research to simulate time series of the climatic parameters from available observational data.

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